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END-SEMESTER EXAMINATION, June 2024

Name of The Program: B.Tech

Name of The Course : Reinforced Cement Concrete - II

Time: 3Hrs.

Note:

- All Questions are Compulsory.
- Answer any two questions among a,b,c in each main question.
- Total Marks for each main question is twenty, each question carries 10 Marks.
- IS codes supplied by the department only are permitted. Any unfair means are warned for action.

Semester : VI semester

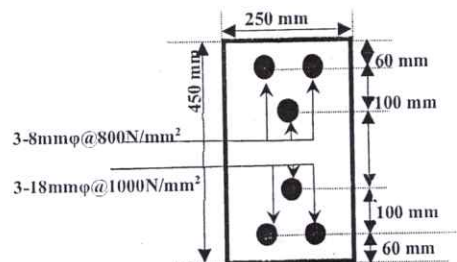
Course Code: TCE 602

Maximum Marks: 100

Q1

2x10 = 20 Marks

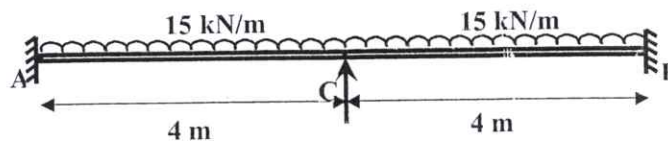
- Compare* the conceptual difference between RCC and PSC philosophy? *Discuss* the advantages and disadvantages of prestressed concrete members? (CO1)
- Analyze* the given prestressed concrete section for extreme fiber stresses at mid span.? (CO1)
Simply supported, rectangular beam section of 200 X 400 mm size, with 8m long span is carrying a UDL of 12kN/m over the entire span and provided with 6-8mm dia steel wires positioned at 75mm from top and bottom fiber of the beam. A net prestress of 950N/mm² is applied on all the wires. Assume Unit Weight of the Concrete as 25kN/m³.
- Inspect* whether the given PSC beam section is concentric or not? (CO1)



Q2

2x10 = 20 Marks

- Figure out* the significance considering various load combinations while analyzing a multi-span continuous beam? (CO2)
- Develop* the bending moment envelop after 20% moment redistribution for the given 2 span continuous beam for maximum negative bending moment at interior support? Consider a characteristic dead load of 5kN/m over entire span. (CO2)



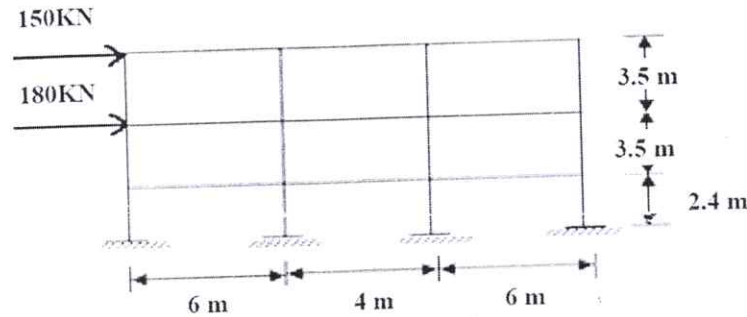
- Illustrate* the benefit of applying the limit analysis concept on a Fixed Beam carrying UDL over the entire span? (CO2)

2x10 = 20 Marks

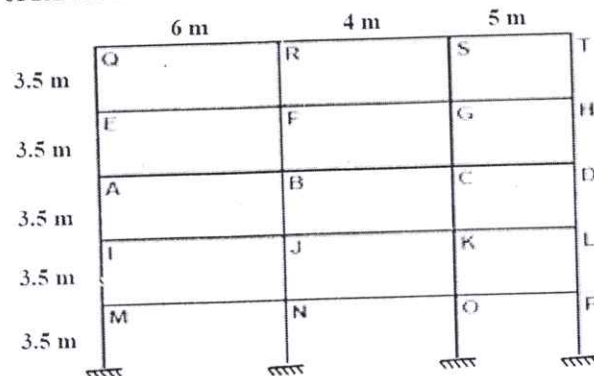
Q3

- (a) **State and explain** the various important assumptions made for 'Approximate Analysis' of a RCC building frames, giving the salient reasoning behind them? (CO3)

- (b) An RCC building frame is given below subjected to horizontal loading as shown in the given figure, such similar frames are at a space of 4.5 m in the transverse direction. Consider all the beams are of size 300mm x 500mm and Column = 450mm x 300mm. Live load on floor = 4kN/m². **Analyze** the frame for the axial forces in the top most storey columns by using cantilever method? (CO3)



- (c) **Analyze** the member forces for the substitute frame EFGH selected, for Max Negative BM at E, if the frame members have same section modulus and load details as given below. Member dead load of 2kN/m and a live load of 4kN/m run. (CO3)



2x10 = 20 Marks

Q4

- (a) Explain the functional **comparative differences** of cantilever and counterfort retaining walls with the help of neat sketches? (CO4)

- (b) **Develop** the soil pressure diagram to be used for the design of a cantilever retaining wall that retain a level difference of 5 meters. The foundation depth may be taken as 0.75 meter from the lower ground level with safe bearing capacity of 180 kN/m². Assume that the backfill has a unit weight of 17 kN/M³ associated with a uniform surcharge of 5 kN/m² on the retention side. The soil has an angle of repose of 35°. Further, assume a coefficient of friction between soil and the base slab as 0.45? (CO4)

- (c) **Design** the stem slab of counter fort retaining wall to retain 7m high embankment above ground level. The foundation must be taken 1.5m deep. Soil SBC is 190kN/m². The retained earth is horizontal and soil unit wt is 15kN/m³ with an angle of internal friction 35°. and coefficient of friction between concrete and soil is 0.40 Use M20 and Fe 415 Grade materials. (CO4)

2x10 = 20 Marks

Q5

- (a) Give a brief *classification* criterion considered while analyzing a water tank? *Highlight* the key considerations and assumptions made while analyzing water tanks on ground and underground ones? (CO5)
- (b) *Design* the long side wall of square concrete water sump of size 4 X 6 m for a capacity of 40,000 liters considering a freeboard of 150mm. The wall of the tank is considered to be fixed at the base and free at the top. Use M25 and Fe 415 steel. (CO5)
- (c) Discuss the sequential steps of designing an elevated water tank components with schematic diagrams? (CO5)

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End Semester Examination 2024Name of the Course: **B.Tech. Civil Engineering**Semester: **VI**Name of the Paper: **Water Resources Engineering II**Paper Code: **TCE-603**

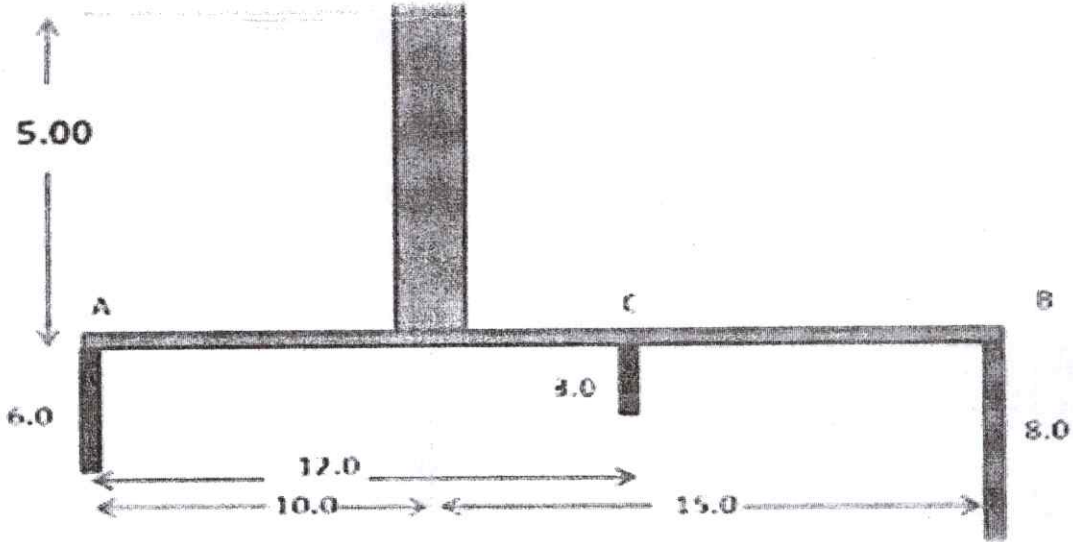
Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each sub question carries 10 marks

Q1	(20marks)	
(a)	Explain briefly with neat sketch's the different forces that may act on a gravity dam.	
(b)	Enumerate different methods of structural stability analysis of gravity dam and explain in brief any one of them.	CO1 & CO6
(c)	A concrete gravity dam has maximum reservoir level as 200.0 m, base level as 115.0 m, top level of dam is 205.0 m and top width as 12.0 m. The upstream face is vertical and the downstream face batter starts at 199.0 m and is 1.0 H : 1.5 V. There is no tail water & consider full uplift pressure at heel. Calculate: i. Maximum vertical stresses at heel and toe. ii. Major principal stress and shear stress near the toe Neglect earthquake and wave action.	
Q2	(20marks)	
(a)	Explain the various criteria for the design of safe earthen dam.	
(b)	Draw a section of an earth dam of 24 m height indicating the various parts of the dam.	
(c)	An earth dam made of a homogeneous material has the following data: Coefficient of permeability of dam material = 5×10^{-4} cm/ sec. Level of top of dam = 200.0 m. Level of deepest river bed = 178.0 m. H.F.L. of reservoir = 197.5 m. Width of the top of dam = 4.5 m. Upstream slope = 3 : 1. Downstream slope = 2 : 1. Horizontal filter of length equal to 25 m is provided inward from the downstream toe of the dam. Determine the phreatic line for this dam section and the discharge passing through per meter length of the earthen dam	CO2 & CO6
Q3	(20marks)	
(a)	What are the different type of spillway and write their suitability.	CO3 & CO6
(b)	Enumerate the various factors affecting the design of energy dissipaters. Give the classification of energy dissipaters based on the position of tail water rating curve w.r.t. jump height curve.	

(c)	A high ogee spillway is to discharge 24 cumecs/m length of spillway. If the maximum water level is to be 50 m above river bed. Obtain the d/s profile of the spillway crest using USWES method, Take the u/s face as vertical and d/s has slope of 0.75H:1.0V. Assuming the coefficient of discharge for ogee spillway as 2.2, rounded piers and rounded abutment.	
Q4	(20marks)	
(a)	Explain the main causes of failure of hydraulic structures founded on pervious foundations. What remedies would you suggest to prevent them?	CO4
(b)	Explain clearly with neat sketches, the various corrections that are to be applied to the pressures at key points, as per Khosla's theory.	
(c)	The figure shows the section of hydraulic structure on permeable foundation. Calculate the average hydraulic gradient by Bligh's creep theory and Lane's creep theory and find uplift pressure and thickness of floor at a distance 10 m. from A (starting point of floor). Also locate the point where required floor thickness is same for both the theory. Take $s = 2.24$ 	
Q5	(20marks)	
(a)	What is meant by water-logging? What are its ill effects, describe some anti-water-logging measure.	CO5
(b)	Discuss the necessity of river training work and enumerate the various works associated with it. Explain any one method with the help of neat sketch.	& CO6
(c)	Determine the spacing of the tile drain from the following data. Average annual rainfall = 800 mm. Discharge usually taken is 1% of average annual rainfall in 24 hours Depth of impervious strata from land surface = 10 m. Highest position of water table below land surface = 1.5 m. Depth of drain = 1.8 m. below land surface. Permeability of soil = 4×10^{-4} m/s.	

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Paper Code: TCE 604

B.Tech (Civil Engineering)
End Semester Examination 2024
VI Semester
Quantity Estimation and Costing

Time: Three Hours

MM: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions in each main question.
- (iii) Total marks for each main question is **Twenty**

Q1

(10X 2 = 20 Marks)

- a) Explain methods of taking out quantities of items of work? Describe various units of measurements in civil engineering works?
- b) Prepare a preliminary estimate of a four storied office building having total carpet Area of 2700.sqm for obtaining the administrative approval of the government, given the following data. It may be assumed that 40% of the built up area will be taken up by corridors, verandah, lavatories, staircase etc. Plinth area rate is Rs. 2225/- per sqm.. Considered the following given charges-
 - (i) Extra for special architectural treatment 0.5% of building cost.
 - (ii) Extra due to deeper foundation at site 1.5% of building cost.
 - (iii) Extra for water supply and sanitary installation 9% of building cost.
 - (iv) Extra for internal electrical installation 14% of building cost.
 - (v) Extra for other services 5% of building cost.
 - (vi) Contingencies - 5%
 - (vii) Supervision charges - 10 %

CO1

- c) Prepare the rough estimate for a proposed commercial complex for a municipal corporation for the following data
 Plinth Area = 450m²/ floor
 Height of each storey = 2.8 m
 No. of storey = G+5
 Cubical content rate = Rupees 1150/m³
 Provided for a following as a percentage of structured cost
 - (i) Water supply & Sanitary arrangement 8%
 - (ii) Electrification works 9%
 - (iii) Architectural appearance 1.5%
 - (iv) Contractors profit 10%
 - (v) Petty supervision & contingencies 5%

Q2

(10X 2 = 20 Marks)

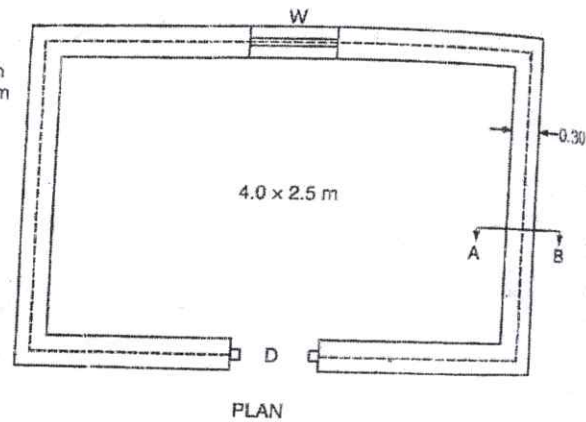
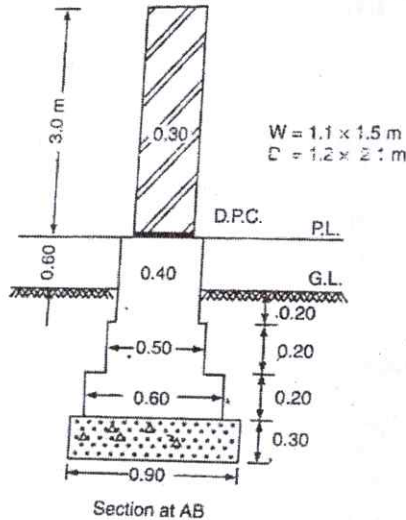
- a) Calculate the quantities of material required for the following items of work:
 - (i) Cement plaster 12 mm thick cement mortar 1:3 for 400m²
 - (ii) Cement concrete 1:3:6 for 10cum
- b) Explain the data required for the analysis of rates?
- c) Calculate the rate for 6mm thick mosaic or terrazzo (1:1) layer over 20mm thick cement concrete flooring for 100sqm assume suitable data

CO2

Q3

(10X 2 = 20 Marks)

- Explain Mid sectional area method for calculating quantity of earthwork in roads
- Estimate by center line method the quantities of the following items of a residential building shown below
 - Earth work in excavation in foundation
 - Lime concrete in foundation(1:7:20)
 - 1st class brickwork in (1:6) cement sand mortar in foundation and up to plinth
 - DPC of 2.5 cm thick of cement concrete 1:2:4



CO3

- Prepare a detailed estimate for earthwork for a portion of a road from the following data:-

Dist. in m	0	100	200	300	400	500	600	700	800	900	1000	1100	1200
R.L. of ground	114.50	114.75	115.25	115.20	116.10	116.85	118.00	118.25	118.10	117.80	117.75	117.90	119.50
R.L. of Formation	115	115	115	115	116	116.85	118.00	118.25	118.10	117.80	117.75	117.90	119.50

1:5 Upward gradient 1 in 200 up to 600 m -- --Downward gradient 1 in 400

Formation width of road is 10 m side slope 2:1 in banking and 1½:1 in cutting. Assume suitable rates

Q4

(10X 2 = 20 Marks)

- Explain the Factors various affecting cost of work in civil engineering works?
- Explain Contingencies and work charge establishment in civil construction works?
- Explain various percentages for different services in Building?

CO4

Q5

(10X 2 = 20 Marks)

- Explain Valuation and purpose of valuation in civil structures works?
- Explain Sinking fund, capital cost, capitalized value?
- Calculate the standard rent of a government residential building newly constructed from the following data—

- Cost of land RS. 25500.00
- Cost of construction of the building Rs 54000.00
- Cost of roads within compound and fencing Rs 1200.00
- Cost of electric installation including fans 7% of the cost of the building.
- Municipal house tax Rs 600.00 per annum
- Water tax Rs 750.00 per annum
- Property tax Rs 880.00 per annum

CO5